Electronic Markets

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Outline

1. Almgren-Chriss Model
2. Finite Markovian Decision Process
3. Back to Almgren-Chriss Model
4. High-frequency trading in a limit order book
5. Policy gradient theorem and REINFORCE
6. Fully recurrent setup

Outline

- Presentation
- About this Course

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5. Policy gradient theorem and REINFORCE

Some words about me

- Jiang PU, lecturer and researcher at ESILV
- Ph.D. in applied mathematics, more specifically in quantitative finance
- Research topic: stochastic optimization, market microstructure, statistical learning
- Contact me by e-mail for questions: jiang.pu.2009@m4x.org

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Courses Objectives

- Provide mathematical tools and modelling for electronic markets
- Understand how trading algorithms work

Grading

- Mainly by project

Recommendation

- This course is quite technical (mathematically)
- You are supposed to work on your own if you do not meet prerequisites:
  - Calculus (especially integral and differential calculus)
  - Ordinary differential equations
  - Stochastic calculus
- Do your homework!
- Do not hesitate to interrupt the class to ask questions
- The slides are just course materials, they are not designed to be comprehensive!!!

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1. Almgren-Chriss Model
   - Introduction
   - Modelling
   - Mean-Variance framework
   - Euler-Lagrange Equation and Solution
2. Finite Markovian Decision Process
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Trading Algorithms

- Most trading algorithm:
  - Strategic layer: controlling risk w.r.t. a benchmark, finding a trading curve (macro)
  - Tactical layer: seeking liquidity inside the order book (micro)
- Strategic layer: Almgren-Chriss (Part I, III)
- Tactical layer: Avellaneda-Stoikov (Part IV)

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Optimal Liquidation

- Large amount Q of a risky asset S to liquidate
- If too fast: liquidity costs
- If too slow: inventory risk

State variables

- Portfolio of risky asset

$$\begin{aligned} q_0 &= Q \\ dq_t &= -v_t dt \end{aligned}$$

- Price

$$dS_t = \sigma dW_t - g(v_t)\, dt$$

Cash

- Cash

$$dX_t = v_t \left(S_t - h\left(v_t\right)\right) dt$$

- X_T?

Payoff

- Implementation Shortfall:

$$X_T - q_0 S_0 = \sigma \int_0^T q_t dW_t - \int_0^T g(v_t) q_t dt - \int_0^T v_t h(v_t) dt$$

Admissible strategies

- Static strategies
- Markovian control

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Mean-variance criterion

- Risk aversion parameter λ
- Criterion to maximize:

$$\mathbb{E}\left[X_T - q_0 S_0\right] - \lambda \mathbb{V}\left[X_T - q_0 S_0\right]$$

- Dimension analysis

Mean-variance optimization

Expressions of $\mathbb{E}\left[X_T - q_0 S_0\right]$ and $\mathbb{V}\left[X_T - q_0 S_0\right]$

$$\begin{aligned}\mathbb{E}\left[X_T - q_0 S_0\right] &= -\int_0^T v_t h\left(v_t\right) dt \\ \mathbb{V}\left[X_T - q_0 S_0\right] &= \sigma^2 \int_0^T q_t^2 dt\end{aligned}$$

Mean-variance optimization

$$\min_{q} \int_0^T \left[-\dot{q}(t)\, h(-\dot{q}(t)) + \lambda \sigma^2 q^2(t) \right] dt$$

$$\text{s.t.} \quad q(0) = Q, q(T) = 0$$

Interpretation

- Almgren-Chriss dynamics + mean-variance optimization with static strategies
- Equivalent to a static optimization problem:
  - Liquidation penalty
  - Inventory penalty

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2. Finite Markovian Decision Process
3. Back to Almgren-Chriss Model
4. High-frequency trading in a limit order book

Euler-Lagrange Equation

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1. Almgren-Chriss Model
2. Finite Markovian Decision Process
   - Framework
   - Policy and value
   - Policy iteration
   - Value Iteration
   - Backward Induction
   - Limitations
3. Back to Almgren-Chriss Model
4. High-frequency trading in a limit order book

Finite Markovian Decision Process

- One agent
- Discrete time: $t = 0, 1, 2, \cdots$
- At step t, the agent receives $S_t \in \mathcal{S}$ (state)
- The agent selects the action $A_t \in \mathcal{A}$

Transition probabilities

- To the next step, he receives the reward $R_{t+1} \in \mathbb{R}$ and finds himself in the state S_{t+1}
- $p(s', r|s, a)$ joint distribution of (S_{t+1}, R_{t+1}) given $S_t = s$, $A_t = a$
- p characterize the environment ("rule", "model")
- Exercise: compute the expected reward given a state/action pair.

Return

- Finite horizon return:

$$\begin{aligned} G_t &= R_{t+1} + \rho R_{t+2} + \cdots + \rho^{T-t-1} R_T \\ &= R_{t+1} + \rho G_{t+1} \end{aligned}$$

- Infinite horizon return (with discount factor ρ):

$$\begin{aligned} G_t &= R_{t+1} + \rho R_{t+2} + \rho^2 R_{t+3} + \cdots \\ &= R_{t+1} + \rho G_{t+1} \end{aligned}$$

- Purpose: choose A_t to maximize $\mathbb{E}[G_t]$

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Policy

- Policy: $\pi : \mathcal{S} \ni S \mapsto A \in \mathcal{A}$
- Or more generally $\pi(a|s)$ distribution of A_t given $S_t = s$

Value

- For a given policy π, the value (performance of the policy) $v_\pi(s) = \mathbb{E}_\pi[G_t | S_t = s]$.
- Consistency condition: for all $s \in \mathcal{S}$,

$$\begin{aligned} v_\pi(s) &= \mathbb{E}_\pi[R_{t+1} + \rho v_\pi(S_{t+1}) | S_t = s] \\ &= \sum_a \pi(a|s) \sum_{s',r} p(s', r|s, a)\left[r + \rho v_\pi(s')\right]. \end{aligned}$$

- State-action value: $q_\pi(s, a) = \mathbb{E}_\pi[G_t | S_t = s, A_t = a]$
- Exercise: relation between q_π and v_π

Bellman equation

- Optimal policy $\pi_\star$ with value $v_\star$ and state-action value $q_\star$
- For all $s \in \mathcal{S}$,

$$\begin{aligned} v_\star(s) &= \max_a \mathbb{E}\left[R_{t+1} + \rho v_\star(S_{t+1}) \,|S_t = s, A_t = a\right] \\ &= \max_a \sum_{s',r} p(s', r|s, a)\left[r + \rho v_\star(s')\right]. \end{aligned}$$

- Exercise: Bellman equation on $q_\star$ and $\pi_\star$

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Policy improvement

- Value: $v_\pi(s) = \mathbb{E}_\pi[G_t | S_t = s]$
- State-action value: $q_\pi(s, a) = \mathbb{E}_\pi[G_t | S_t = s, A_t = a]$
- Room for improvement, when $q_\pi(s, a) > v_\pi(s)$ for some (s, a)
- Policy improvement theorem: if π' satisfies $q_\pi(s, \pi'(s)) \geq v_\pi(s)$ for all s, then $v_{\pi'}(s) \geq v_\pi(s)$ for all s

Policy iteration

- Initialize value function V and policy π arbitrarily
- Loop until no further modification of the policy:
  - Policy evaluation using the consistency condition
  - Policy improvement using the policy improvement theorem

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Value Iteration

- Initialize value function V arbitrarily
- Loop:
  - value iteration using Bellman's equation
- Obtain policy π (at the same time using Bellman's equation)

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4. High-frequency trading in a limit order book

Backward Induction

- Special case: irreversible
- Initialize value function V for terminal nodes
- Loop through backward propagation
  - Obtain policy π at the same time

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4. High-frequency trading in a limit order book

Limitations

- $\mathcal{S}$ too big
- $\mathcal{A}$ too big
- p not known
- Markovian assumption
- Other objective functions

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3. Back to Almgren-Chriss Model
   - Expected utility framework framework
   - CARA utility + static strategy
   - CARA utility + Markovian control
   - Hamilton-Jacobi-Bellman Equation
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4. High-frequency trading in a limit order book

Expected utility framework

- Random payoff Y
- Agent with utility function u
- $\max \mathbb{E}\left[u\left(Y\right)\right]$ instead of $\mathbb{E}\left[Y\right]$

Certainty equivalent

- Certainty equivalent $\mathrm{CE}(Y)$: the number you put such that $u(\cdot) = \mathbb{E}[u(Y)]$
- Maximizing $\mathrm{CE}(Y)$ is equivalent to maximizing $\mathbb{E}[u(Y)]$

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CARA utility functions

- $u : x \mapsto -\exp(-\gamma x)$
- $\mathrm{CE}(Y + a) = \mathrm{CE}(Y) + a$ for deterministic as
- What if Y is Gaussian?

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Markovian decision process

- Like in the previous section (except it is not finite)
- (Optimal) value function

$$v(t,q,s,x) = \sup_{(v_s)_{s\in[t,T]}} \mathbb{E}_t\left[u(X_T)\,|q_t = q, S_t = S, X_t = x\right]$$

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HJB equation

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Dimension Reduction

Interpretation

- Indifference price
- Solution to the simpler 2D problem (without randomness) gives us the solution to the 4D problem

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Market making using limit orders

- Avellaneda & Stoikov (2008)
- Market making

Limit order books

- Underlying price S

$$dS_t = \sigma dW_t$$

- Limit orders p^b and p^a

$$\begin{aligned} \delta^b &= S - p^b \\ \delta^a &= p^a - S \end{aligned}$$

- Market buy (resp. sell) order arrivals N^a (resp. N^b) with Poisson rate $\lambda^a(\delta^a)$ (resp. $\lambda^b\left(\delta^b\right)$)

Inventory

- Cash

$$dX_t = p^a dN_t^a - p^b dN_t^b$$

- Stock (discrete)

$$dq_t = dN_t^b - dN_t^a$$

Next?

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Infinite Markovian Decision Process

- One agent
- Discrete time: $t = 0, 1, 2, \cdots$
- At step t, the agent receives $S_t \in \mathcal{S}$ (can be infinite)
- The agent selects the action $A_t \in \mathcal{A}$ (can be infinite)

Policy

- Policy: $\pi : \mathcal{S} \ni S \to A \in \mathcal{A}$
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2. Finite Markovian Decision Process
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Bellman equation on π

- If π is optimal, π satisfies the Bellman equation

$$\pi(s) \quad \in \quad \arg\max_a \mathbb{E}\left[R_{t+1} + \rho v_\pi(S_{t+1}) \,|S_t = s, A_t = a\right]$$

- Policy improvement?

$$\pi(s) \quad \leftarrow \quad \arg\max_a \mathbb{E}\left[R_{t+1} + \rho v_\pi(S_{t+1}) \,|S_t = s, A_t = a\right]$$

- Not really applicable

Functional approximation

- $\pi_\theta : \mathcal{S} \ni S \mapsto A \in \mathcal{A}$, where $\theta \in \mathbb{R}^d$
- $\pi_\theta(\cdot|s)$ distribution taking values in $\mathcal{A}$, chosen from a family parameterized by a function of s

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Policy gradient theorem

Family of stochastic policies $\left(\pi^{\theta}\right)_{\theta}$ (assume $\rho = 1$):

Policy gradient theorem (to be continued)

$$\begin{aligned}
\nabla_{\theta} v_{\pi^{\theta}}(s_0) &\propto \sum_{s} \mu_{s_0,\pi^{\theta}}(s) \sum_{a} \nabla_{\theta} \pi^{\theta}(a|s)\, q_{\pi}(s,a) \\
&= \mathbb{E}_{\pi^{\theta}}\left[\frac{1}{T} \sum_{t} \sum_{a} q_{\pi^{\theta}}(S_t, a)\, \nabla_{\theta} \pi^{\theta}(a|S_t)\right] \\
&= \mathbb{E}_{\pi^{\theta}}\left[\frac{1}{T} \sum_{t} q_{\pi^{\theta}}(S_t, A_t)\, \nabla_{\theta} \log \pi^{\theta}(A_t|S_t)\right]
\end{aligned}$$

Policy gradient theorem

... Policy gradient theorem

We have also

$$\begin{aligned} & \mathbb{E}_{\pi^\theta}\left[\frac{1}{T}\sum_t q_{\pi^\theta}\left(S_t, A_t\right)\nabla_\theta \log \pi^\theta\left(A_t|S_t\right)\right] \\ = \quad & \mathbb{E}_{\pi^\theta}\left[\frac{1}{T}\sum_t G_t \nabla_\theta \log \pi^\theta\left(A_t|S_t\right)\right] \end{aligned}$$

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REINFORCE

- Loop
  - Simulate the current policy π^θ: S_0, A_0, R_1, S_1, A_1, R_2, $\cdots$, S_{T-1}, A_{T-1}, R_T
  - For $t = 0, \cdots, T-1$:
    - $\theta \leftarrow \theta + \alpha G_t \nabla_\theta \log \pi^\theta (A_t | S_t)$

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Adding a baseline (variance reduction)

- Policy gradient theorem:
  $\nabla_\theta v_{\pi^\theta}(s_0) \propto \mathbb{E}_{\pi^\theta}\left[G_t \nabla_\theta \log \pi^\theta(A_t|S_t)\right]$
- Still holds if modified with φ that does not depend on the action chosen:
  $\nabla_\theta v_{\pi^\theta}(s_0) \propto \mathbb{E}_{\pi^\theta}\left[(G_t - \varphi(t, S_t))\nabla_\theta \log \pi^\theta(A_t|S_t)\right]$
- Variance reduction if $\varphi(t, S_t) \approx v_{\pi^\theta}(S_t)$

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   - Fully recurrent setup

Fully recurrent setup

- Initial state S_0, deterministic functions R and P and noises $\left(\xi^R\right)$ and $\left(\xi^S\right)$
- Agent performs action $A_1 = \pi^\theta\left(S_0\right)$, given by the deterministic policy π^θ
- $R_1 = R\left(S_0, A_1, \xi_1^R\right)$, $S_1 = P\left(S_0, A_1, \xi_1^S\right)$
- Agent performs action $A_2 = \pi^\theta\left(S_1\right)$
- $R_2 = R\left(S_1, A_2, \xi_2^R\right)$, $S_2 = P\left(S_1, A_2, \xi_2^S\right)$

...

Dependency on θ

- Given the noises $\left(\xi^R\right)$ and $\left(\xi^S\right)$, S_t is known.
- $S_t = P\left(S_{t-1}, A_{t-1}, \xi_t^S\right)$ depends on θ through P's second argument: $A_{t-1} = \pi^\theta\left(S_{t-1}\right)$, but also through P's first argument by induction
- Same for R_t
- To maximize $\mathbb{E}\left[\sum_t R_t\right]$:
  - Use Monte Carlo by sampling $\left(\xi^R\right)$ and $\left(\xi^S\right)$
  - differentiate w.r.t. θ, given $\left(\xi^R\right)$ and $\left(\xi^S\right)$, by taking into account ALL the dependencies on θ.
    $\rightarrow$Fully recurrent neural network, then backpropagate
- Also possible to use mean-variance criterion, expected utility